CrowdWisdomTrading X Sentiment Price Prediction agent
Intern position assessment

The document gives a brief introduction to an assignment to check your qualifications for the internship as a Quantitative Data Scientist. Please complete it on your own time and submit it within 1 week.

Project Overview

Create a Python-based machine learning pipeline that predicts an asset's price at n+0, n+1, n+2, n+3, n+4 and n+5 based on X (Twitter) sentiment and price only. The system will scrape X posts, ingest recent news, join the datasets in a database, and train a predictive model using rigorous out-of-sample validation.

Technical Requirements

https://apify.com/ - for data scraping (alternative data features)

Project Scope

Develop X sentiment price prediction model

1. X Sentiment & Price Ingestion (Apify-based)
   - Scrape X (Twitter) posts about the chosen asset (ticker, cashtag and keyword searches) for the last 180 days using Apify, plus historical price data for the same period.
   - Extract post timestamps (UTC/EST alignment), post text, engagement (likes, reposts, views) and author follower counts, and align them with the price bars.
   - most recent and relevant news on the asset to complement the X sentiment (use tavily.com)
2. Quantitative Feature Engineering & Sentiment summary
   - Feature Engineering: Create time-based features from sentiment and price only (e.g. sentiment score, post volume, lagged returns, time horizon)
   - 1 page summary of the current sentiment outlook for the asset
3. Machine Learning Model & Walk-Forward Validation
   - Build a predictive model that uses sentiment and price only to forecast the price at n+0, n+1, n+2, n+3, n+4 and n+5, where n is the current period.
4. Visual Matrix Output Generator
   - Charts that can visual the X sentiment change against price as time changes

Submission by email gilad@crowdwisdomtrading.com **including apify/tavily/exa**

Submit the following:

1. A link to your GitHub/GitLab repository.
2. APIFY /tavily/exa tokens and actor IDs used for scraping (so we can reproduce your pipeline).
3. An evaluation report (PDF or Markdown) detailing model performance metrics (Sharpe Ratio, Sortino Ratio, Max Drawdown).

Timeline

You will have 5 - 7 days to complete this task. Please ensure you submit the deliverables by the end of this period.

We look forward to seeing your solution\!

Thanks for applying!
Gilad
CrowdWisdomTrading CEO